

Student Assignment Brief

This document is for CU Group students for their own use in completing their assessed work for this module and should not be passed to third parties or posted on any website. If you require this document in an alternative format, please contact your Module Tutor.

Assignment Information		
The work you submit for this assignment must be your own independent work, or in the case of group assignment your own groups' work. More information is available in the 'Assessment Task' section of this assignment brief.		
Module Name:	Security	
Module Code:	NB6005CEM	
Assignment Title	System Security Evaluation (Module Learning Outcome 1 and 4)	
Assignment Due:	xxxxxxxxxx	
Assignment Type: Only one of the following apply to this assignment task.	☐ Pass or Fail (Core Assessment) You will be provided with either a pass or a fail grade. You will have to successfully complete your assessment before the end of the module.	☒ Percentage Grade (Applied Core Assessment) You will be provided with an overall grade between 0% and 100%. You have one opportunity to pass the assignment at or above 40%.

Assignment Tasks	
- You are developing a secure banking application. - Draw a system design diagram - No need to go to database design level - ER diagrams etc. Take the database as one unit, and draw the rest of the parts in the systems - network, switches etc.) (20 %) - Indicate on the diagram where security can be applied to the system indicating the potential vulnerabilities (Access Control points, Firewall points, Encryption points/ level etc.) (40 %) - Describe all the security mechanisms you used in the system at algorithm level (Encryption/ Decryption - Symmetric Key and Asymmetric Key , hashing, digital signatures etc.) protocol level and system level at each point you described in Part b. (40 %) Write technical report of the system containing: An Abstract (20 %) Security Analysis of the system , incorporating Part 1.b and Part 1.c (80 %)	
Submission Instructions: Submit to the link in the LMS.	

Marking and Feedback

Your assignment will be marked . Provisional grades will be released . Feedback will be provided .

Feedback can be accessed in the respective submission link on LMS.

Details of the marking criteria for this task can be found at the bottom of this assignment brief.

Module Learning Outcomes:

The Learning Outcomes for this module align to the **marking criteria** which can be found at the end of this brief. Ensure you understand the marking criteria to ensure successful achievement of the assessment task.

The following module learning outcomes are assessed in this task:

1. Critically evaluate a range of encryption and authentication methods for a given set of requirements.
2. Utilise systematic knowledge to create secure environments at the host or network level.
3. Develop and evaluate software that addresses the most common and most severe security concerns.
4. Critically evaluate the security of an IT ecosystem.

Assignment Support and Academic Integrity

Late Submission

If you are not able to complete your coursework on time due to extenuating circumstances*, the ONLY way to receive an extension (up to 5 working days) or a deferral (anything longer than 5 working days) is to apply using the [online portal](#) where you will also be able to find more information about extensions and deferrals any evidence required.

*Extenuating circumstances are defined by CU as 'genuine circumstances beyond your control or ability to foresee, and which seriously impair your assessed work'. Please note that you will need to provide third party evidence to support your reasoning for requiring an extension or deferral. Your course tutor is NOT able to approve an extension or a deferral. If you have not completed the official forms, your work will count as not submitted and receive a zero mark.

Spelling, Punctuation, and Grammar:

You are expected to use effective, accurate, and appropriate language within this assessment task.

Academic Integrity:

The work you submit must be your own work. We use detection software and make routine checks for evidence of academic misconduct. Definitions of academic misconduct, including plagiarism, self-plagiarism, and collusion can be found [on the Student Portal](#). All cases of suspected academic misconduct are referred for investigation, the outcomes of which can have profound consequences to your studies. For more information on academic integrity please visit the [Academic and Research Integrity](#) section of the Student Portal.

Support for Students with Disabilities or Additional Needs:

If you have a disability, long-term health condition, specific learning difference, mental health diagnosis or symptoms and have discussed your support needs with health and wellbeing you may be able to access support that will help with your studies.

If you feel you may benefit from additional support, but have not disclosed a disability to the University, or have disclosed but are yet to discuss your support needs it is important to let us know so we can provide the right support for your circumstances. Visit [the Student Portal](#) to find out more.

Guidance Notes and Considerations

Referencing

Access, Foundation and Undergraduate

You should use APA 7th edition style of referencing.

Referencing guidance can be accessed on the Library's [LibGuides pages](#).

Law

You should use OSCOLA referencing.

Referencing guidance can be accessed on the Library's [LibGuides pages](#).

Retakes or deferrals where CU Harvard referencing was previously used

If you have previously used the CU Harvard style of referencing, you can continue to use this until you graduate. Although you can choose to make the switch to APA at any time, you must use a single consistent style of referencing within each assessment.

Referencing guidance can be accessed on the Library's [LibGuides pages](#).

Use of Sources and Information

You can access high quality, authoritative information via the library. Use [LibGuides](#) to access Locate, [BibliU](#) and research guidance.

Academic Writing Resources

There are a variety of academic writing resources available which can be accessed via [LibGuides](#). These writing guides can help with different types of assessment as well as important writing skills needed for university.

[Glossary of Assessment Terms for Assessment Writing](#) – this is an a-z that covers the majority of terms used in assessment briefs, learning outcomes and feedback throughout the CU Group. If you would like any further support with your assessment, you can contact your Academic Writing Developer or visit the Academic Writing [LibGuides](#) page.

If you would like to book an Academic Writing appointment for support with your assessment, you can contact your Writing Developers:

CUC: learningservices.cuc@coventry.ac.uk

CUS: academicskills.cus@coventry.ac.uk

CUL Dagenham: academicwritingservice@cul.coventry.ac.uk

CUL Greenwich: awsgreenwich.cul@coventry.ac.uk

Academic Integrity Guidance

The best way to avoid academic misconduct is to follow appropriate academic and referencing conventions.

Further guidance on academic integrity and conduct can be found using [LibGuides](#).

Collusion between students (where sections of your work are similar to the work submitted by other students in this or previous module cohorts) is taken extremely seriously and will be reported to the Academic

Conduct Panel. This applies to all coursework and exam answers. If you would like more guidance on understanding collusion, you can find it on [LibGuides](#).

If an assessment suspected of involving a breach of academic integrity is found to display a marked difference in writing style, knowledge and skill level from that demonstrated elsewhere on the course, you may be required to undertake a Viva Voce in order to prove the coursework assessment is entirely your own work.

Proofreading of assessments by CU approved proofreaders is permitted. There is a list of [approved proof readers](#) along with [guidelines](#) for use. However, please remember that proofreading is a lengthy and detailed process for which there is a cost. If you decide to use a proofreader, please take this into account and contact them at least 10 days in advance of your assessment deadline.

You must not submit work for an assessment that you have already submitted (partially or in full), either for your current course or for another qualification of this university, unless this is specifically provided for in your assessment brief or specific course or module information.

It is important to realise that as a student you should not submit all or part of an assessment for which you have already received academic credit, to be used for an assessment in a different module. Reusing your own work in this way is called self-plagiarism. Where you wish to refer to some of your own work you must reference it in the same way that you reference work by other people.

Coventry University Generic Assessment Criteria: Undergraduate

Mark band	Outcome	Guidelines
80-100% 1st	Meets learning outcomes	1st - Exceptional work with very high degree of understanding, creativity, and critical/analytic skills. Evidence of exceptional research well beyond minimum recommended using a range of methodologies. . Exceptional understanding of knowledge and subject-specific theories. Demonstrates creative flair, a high degree of originality and autonomy. Exceptional ability to apply learning resources. Demonstrates well-developed problem-solving skills. Work completed with very high degree of accuracy and proficiency and autonomy. Exceptional communication and expression, significant evidence of professional skill set. Student evidences deployment of a full range of exceptional technical, including proficiency in the English Language, and/or artistic skills.
70-79% 1st		1st - Excellent work with clear evidence of understanding, creativity and critical/analytical skills. Thorough research well beyond the minimum recommended using methodologies beyond the usual range. Excellent understanding of knowledge and subject-specific theories with evidence of considerable originality and autonomy. Excellent ability to apply learning resources. Demonstrates consistent, coherent substantiated argument and interpretation. Demonstrates considerable creativity and clear problem-solving skills. Assessment completed with accuracy, proficiency, and considerable autonomy. Excellent communication and expression, some evidence of professional skill set. Student evidences deployment of a highly developed range of technical, including proficiency in the English Language, and/or artistic skills.
60-69% 2:1		2:1 - Very good work demonstrating strong understanding of theories, concepts and issues with clear critical analysis. Thorough research, using established methodologies accurately, beyond the recommended minimum with little, if any, irrelevant material present. Very good understanding, evidencing breadth and depth, of knowledge and subject-specific theories with some originality and autonomy. Very good ability to apply learning resources. Demonstrates coherent substantiated argument and interpretation. Demonstrates some originality, creativity and problem-solving skills. Work completed with accuracy, proficiency, and autonomy. Very good communication and expression with evidence of professional skill set. Student has a thorough command of a good range of technical, including proficiency in the English Language, and/or artistic skills.

50-59%		2:2 - Good understanding of relevant theories, concepts and issues with some critical analysis. Research undertaken accurately using established methodologies, enquiry beyond that recommended may be present. Some errors may be present and some inclusion of irrelevant material. Good understanding, with evidence of breadth and depth, of knowledge and subject-specific theories with indications of originality and autonomy. Good ability to apply learning resources. Demonstrates logical argument and interpretation with supporting evidence. Demonstrates some originality, creativity and problem-solving skills but with inconsistencies. Expression and presentation mostly accurate, proficient, and conducted with some autonomy. Good communication and expression with appropriate professional skill set. Student consistently demonstrates a well-developed range of technical, including proficiency in the English Language, and/or artistic skills.
40-49%		3rd - Meet the learning outcomes with a basic understanding of relevant theories, concepts and issues. Demonstrates an understanding of knowledge and subject-specific theories sufficient to deal with concepts. Assessment may be incomplete and with some errors. Research scope sufficient to evidence use of some established methodologies. Some irrelevant material likely to be present. Basic ability to apply learning resources. Demonstrates ability to devise and sustain an argument. Demonstrates some originality, creativity and problem-solving skills but with inconsistencies. Expression and presentation sufficient for accuracy and proficiency. Sufficient communication and expression with basic professional skill set. Student demonstrates technical, including a basic level of proficiency in the English Language, and/or artistic skills.
30-39%	Fails to achieve learning outcomes	Fail – Outcomes not met. Limited understanding of relevant theories, concepts and issues. Little evidence of research and use of established methodologies. Some relevant material will be present. Deficiencies evident in analysis. Fundamental errors and some misunderstanding likely to be present. Limited ability to apply learning resources. Student’s arguments are weak and poorly constructed. Limited originality, creativity, and struggles with problem-solving skills. Expression and presentation insufficient for accuracy and proficiency. Insufficient communication and expression and with deficiencies in professional skill set. Student demonstrates some deficiencies in technical, including in their use of the English Language, and/or artistic skills.
0-29%		Fail – Outcomes not met. Clear failure demonstrating very little understanding of relevant theories, concepts and issues. Minimal evidence of research and use of established methodologies and incomplete knowledge of the area. Serious and fundamental errors and aspects missing. Very little evidence of ability to apply learning resources. Student’s arguments are very weak and with no evidence of alternative views. Little evidence of originality, creativity, and problem-solving skills. Expression and presentation

		deficient for accuracy and proficiency. Insufficient communication and expression and with deficiencies in professional skill set. Student demonstrates a lack of technical, including in their use of the English Language, and/or artistic skills.
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